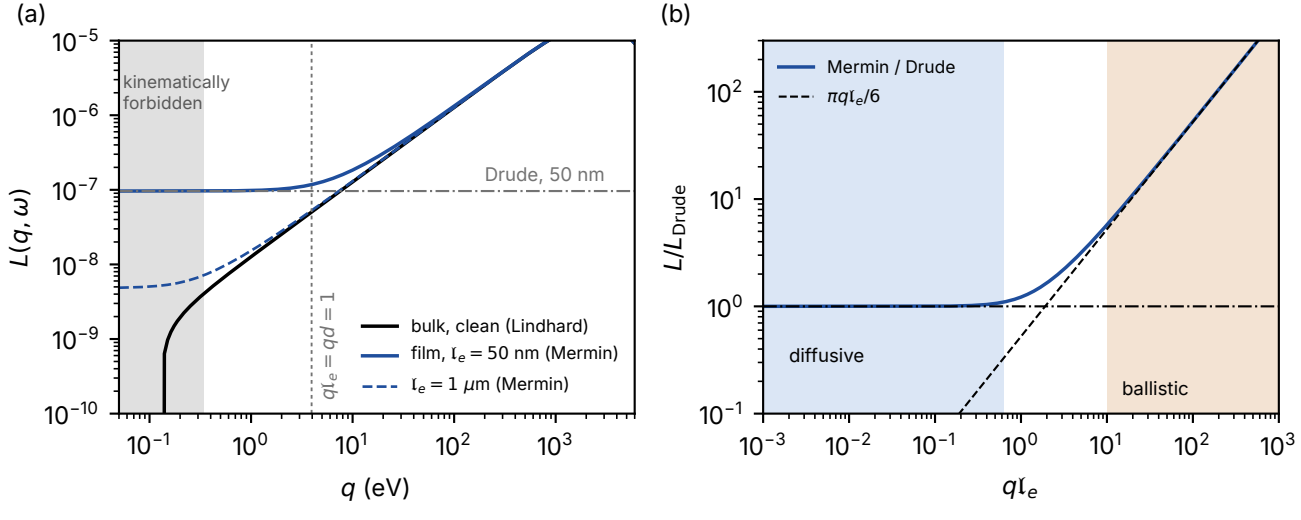
(a)
kinematically forbidden
Drude, 50 nm
$q\ell_e = qd = 1$
bulk, clean (Lindhard)
film, $\ell_e = 50$ nm (Mermin)
$\ell_e = 1$ μm (Mermin)
$L(q,\omega)$
q (eV)
(b)
Mermin / Drude
$\pi q\ell_e/6$
diffusive
ballistic
L/L_{Drude}
$q\ell_e$